

Executive Summary

This project analyzes sales data to identify high-performing product categories, profitable regions, customer segments, and the impact of discounts on profitability. The analysis was conducted using Python, Pandas, Matplotlib, and Seaborn.

The findings reveal that Office Supplies generated the highest sales and profit, while Furniture showed lower profitability. The Consumer segment contributed the highest revenue, and sales peaked during the final months of the year. The study also found a negative relationship between discounts and profit, indicating that excessive discounting reduces profitability.

Based on the analysis, recommendations include optimizing discount strategies, focusing on high-performing categories, strengthening customer retention, and expanding business efforts in underperforming regions.

Dataset Overview

- Dataset: Global Superstore Sales Dataset
- Total Records: 51,290
- Features: 21
- Missing Values: 0
- Tools Used: Python, Pandas, Matplotlib, Seaborn

Sales Performance Analysis – Visualizations & Insights

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
df = pd.read_csv(
    r"D:\DA INTERN\da vs\Sales performance analysis\SuperStoreOrders.csv"
)

df.head()
```

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state	country	market	region	product_id	category	sub_category	product_name
0	AG-2011-2040	1/1/2011	6/1/2011	Standard Class	Toby Braunhardt	Consumer	Constantine	Algeria	Africa	Africa	OFF-TEN-10000025	Office Supplies	Storage	Tenex Lock B
1	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	OFF-SU-10000618	Office Supplies	Supplies	Acme Trimm High Spe
2	HU-2011-1220	1/1/2011	5/1/2011	Second Class	Annie Thurman	Consumer	Budapest	Hungary	EMEA	EMEA	OFF-TEN-10001585	Office Supplies	Storage	Tenex B Single Wic
3	IT-2011-3647632	1/1/2011	5/1/2011	Second Class	Eugene Moren	Home Office	Stockholm	Sweden	EU	North	OFF-PA-10001492	Office Supplies	Paper	Enermax No Car Premi
4	IN-2011-47883	1/1/2011	8/1/2011	Standard Class	Joseph Holt	Consumer	New South Wales	Australia	APAC	Oceania	FUR-FU-10003447	Furniture	Furnishings	Eldon Liç Bulb, Duo Pi

1. Sales by Category

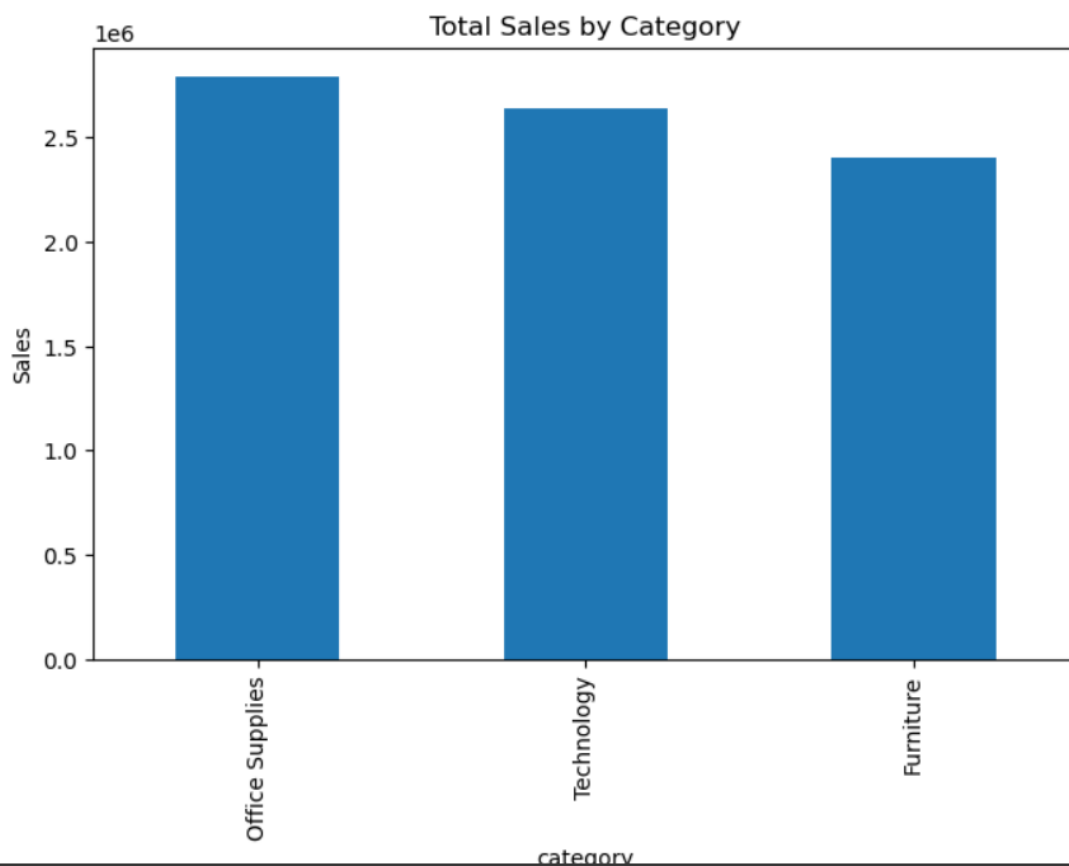
```
] category_sales = df.groupby('category')['sales'].sum().sort_values(ascending=False)

print(category_sales)

plt.figure(figsize=(8,5))
category_sales.plot(kind='bar')

plt.title("Total Sales by Category")
plt.ylabel("Sales")
plt.show()
```

```
category
Office Supplies    2790258.0
Technology         2638265.0
Furniture          2406605.0
Name: sales, dtype: float64
```



2. Profit by Category

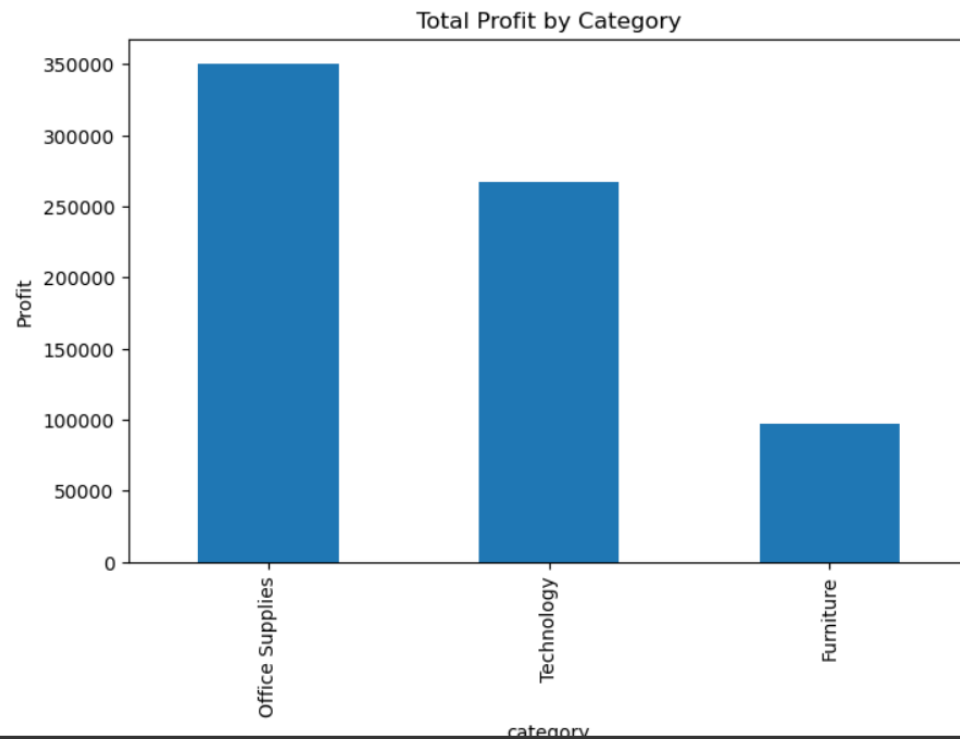
```
15]: category_profit = df.groupby('category')['profit'].sum().sort_values(ascending=False)

print(category_profit)

plt.figure(figsize=(8,5))
category_profit.plot(kind='bar')

plt.title("Total Profit by Category")
plt.ylabel("Profit")
plt.show()
```

```
category
Office Supplies    350107.32450
Technology         267573.47238
Furniture          97049.37790
Name: profit, dtype: float64
```



3. Sales by Region

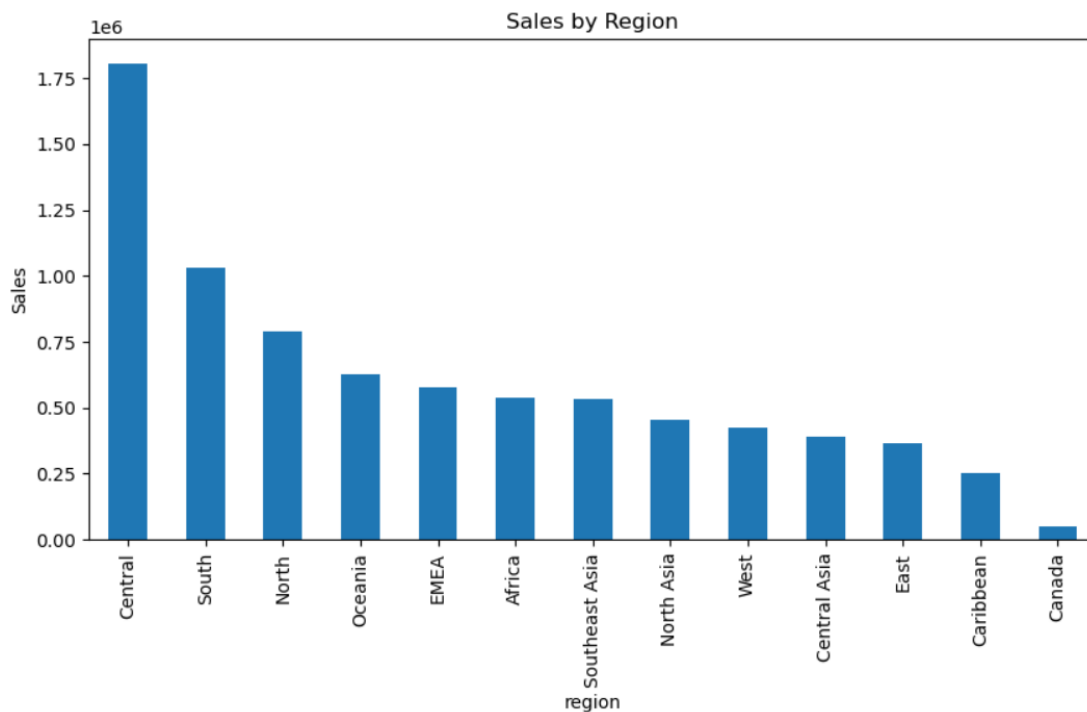
```
16]: region_sales = df.groupby('region')['sales'].sum().sort_values(ascending=False)

print(region_sales)

plt.figure(figsize=(10,5))
region_sales.plot(kind='bar')

plt.title("Sales by Region")
plt.ylabel("Sales")
plt.show()
```

```
region
Central      1806638.0
South        1031101.0
North         790546.0
Oceania       625382.0
EMEA          575562.0
Africa        538115.0
Southeast Asia 532172.0
North Asia    453686.0
West          424173.0
Central Asia  389506.0
East          366492.0
Caribbean    251441.0
Canada        50314.0
Name: sales, dtype: float64
```



4. Top 10 Products by Sales

```
] : top_products = df.groupby('product_name')['sales'].sum().sort_values(ascending=False).head(10)

print(top_products)

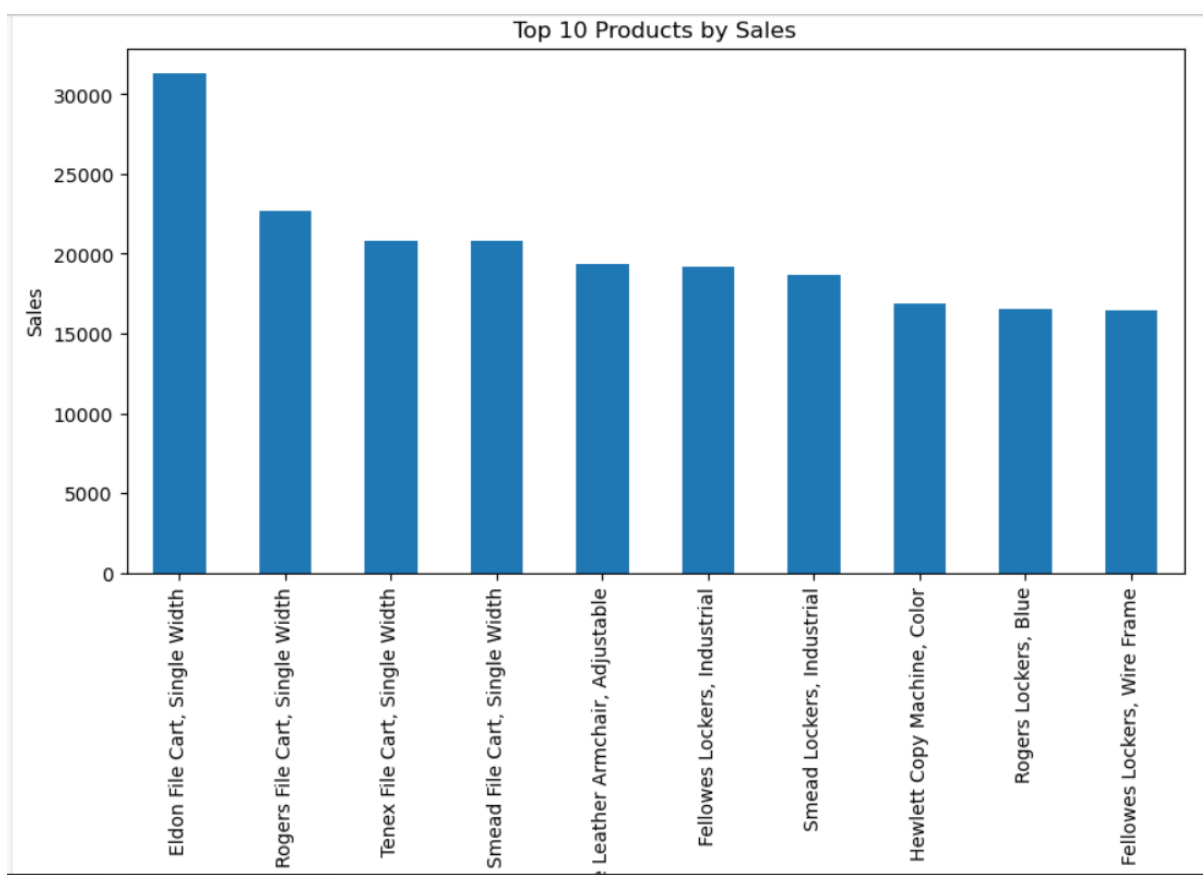
plt.figure(figsize=(10,5))
top_products.plot(kind='bar')

plt.title("Top 10 Products by Sales")
plt.ylabel("Sales")
plt.show()
```

```

product_name
Eldon File Cart, Single Width      31319.0
Rogers File Cart, Single Width     22645.0
Tenex File Cart, Single Width      20778.0
Smead File Cart, Single Width      20775.0
Office Star Executive Leather Armchair, Adjustable  19355.0
Fellowes Lockers, Industrial       19172.0
Smead Lockers, Industrial          18648.0
Hewlett Copy Machine, Color       16849.0
Rogers Lockers, Blue               16494.0
Fellowes Lockers, Wire Frame      16470.0
Name: sales, dtype: float64

```



5. Discount vs Profit Analysis

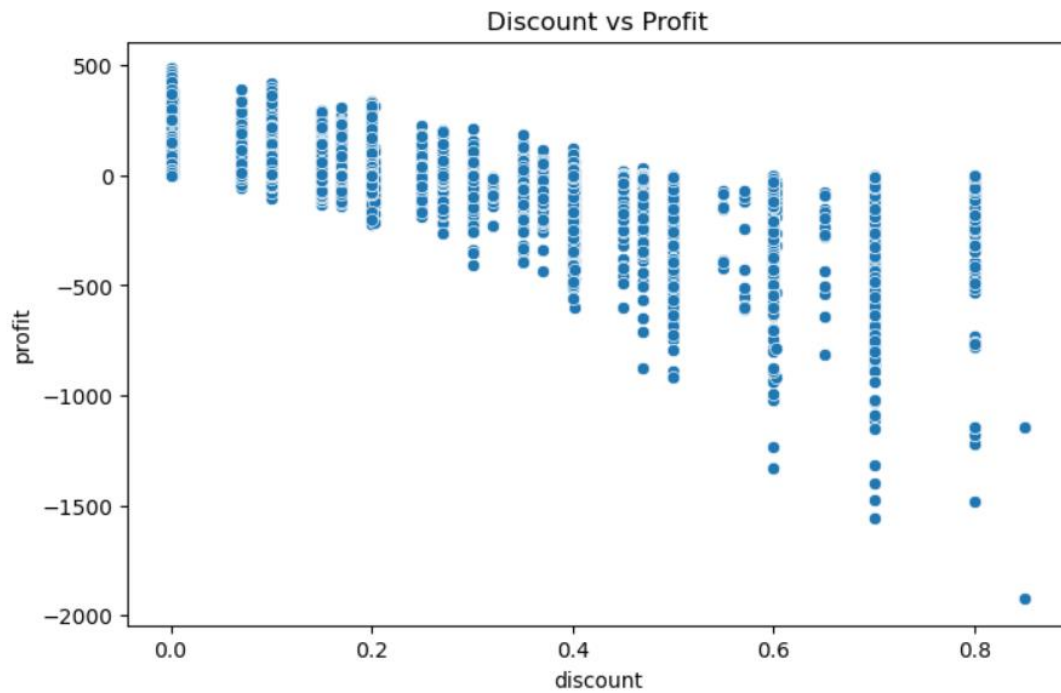
```

3]: plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x='discount',
    y='profit'
)

plt.title("Discount vs Profit")
plt.show()

```



6. Monthly Sales Trend

```
19]: df['month'] = df['order_date'].dt.month

monthly_sales = df.groupby('month')['sales'].sum()

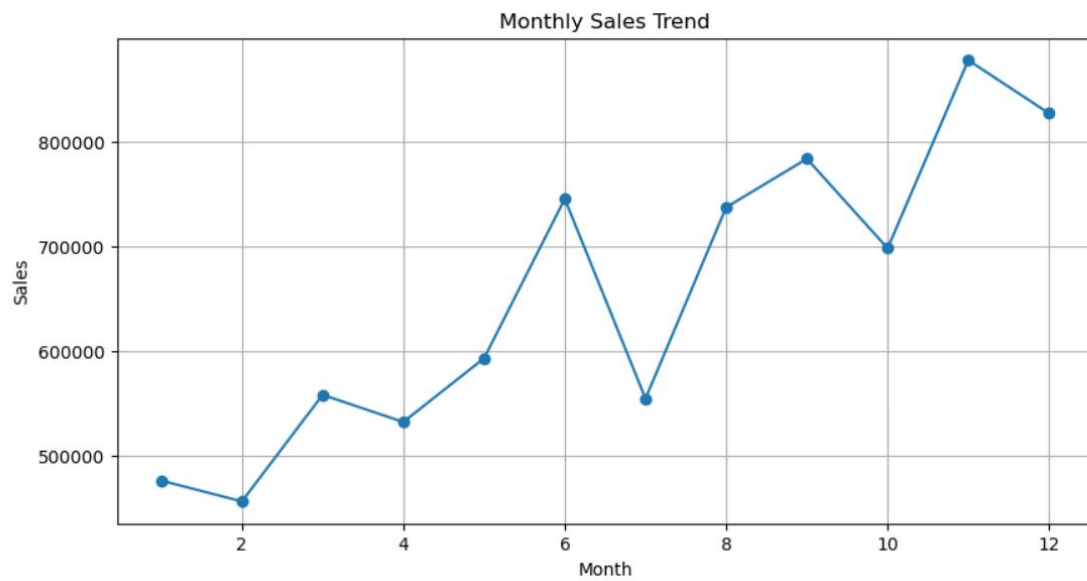
plt.figure(figsize=(10,5))

monthly_sales.plot(marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



7. Customer Segment Analysis

```
|: segment_sales = df.groupby('segment')['sales'].sum()

print(segment_sales)

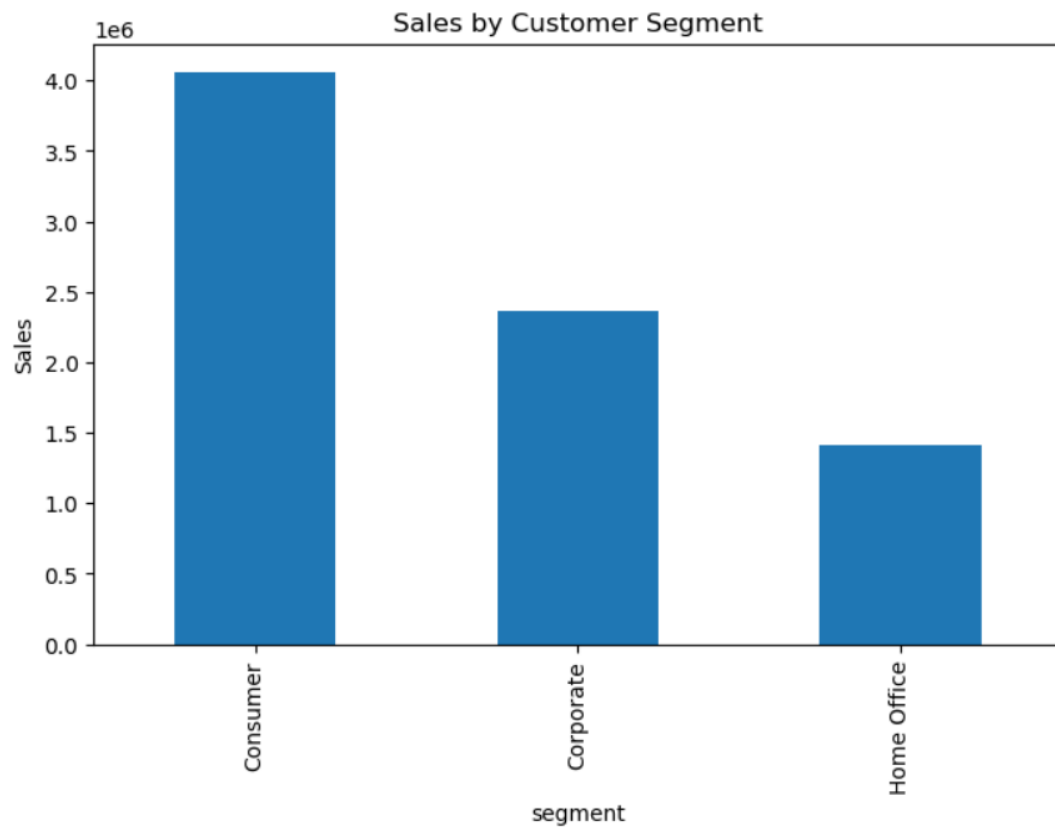
plt.figure(figsize=(8,5))

segment_sales.plot(kind='bar')

plt.title("Sales by Customer Segment")
plt.ylabel("Sales")

plt.show()
```

```
segment
Consumer      4058118.0
Corporate      2369261.0
Home Office    1407749.0
Name: sales, dtype: float64
```



Key Findings

Category Performance

- Office Supplies generated the highest sales and profit.
- Technology performed strongly across all metrics.
- Furniture had the lowest profitability.

Regional Performance

- Central region recorded the highest sales.
- Canada and Caribbean showed the lowest sales.

Customer Segment Analysis

- Consumer customers generated the highest revenue.
- Home Office customers generated the lowest revenue.

Discount Analysis

- Higher discounts reduced profitability.
- Several highly discounted transactions generated losses.

Sales Trend Analysis

- Sales increased throughout the year.
- Peak sales occurred during November and December.

Recommendations

1. Focus on Office Supplies and Technology products.
2. Optimize Furniture pricing and operational costs.
3. Reduce excessive discounting.
4. Expand business activities in weak-performing regions.
5. Strengthen customer retention programs for Consumer customers.

GITHUB LINK :- <https://github.com/TejasSinha2006/Sales-Performance-Analysis>